This project investigates the BRFSS-type dataset related to health issues that includes age, current weight (weight2), weight one year ago (wtyrago), height (htm3), and sex (code: 1=male, 2=female). These variables are important because it indicate trends related to the size and growth and of adults and annual changes. The objective is to create a comprehensive computer analysis system of data cleaning, analysis, graphing, correlation analysis, hypothesis tests, and a discussion related to the issue of ethics. This submission meets the requirements of the assignment which include data cleaning, Task 1, Task 2, Task 3, the discussion of ethics, and the conclusion. In addition, I also tried to understand the patterns to the lives of real people. The data of a person's body weight could mean dietary practices, physiological conditions, stress levels, and lifestyle factors. The privacy issues of biases related to the information of a person's weight are also addressed at the end of this document.

The first step is to develop a personalized sample using the Student ID. I used the last digits of my Student ID to make a specific dataset, for which I would get the same rows every time after running the code. Then I kept the required five columns since the brief required us to concentrate on the five variables only. While checking the missing data, I noticed that I had 454 missing ages, 1,847 missing data for weight2, 2,800 missing data for wtyrago, and 650 missing data for heights. The only feature that lacked any missing data was the sex feature. Although the original data has 50,000 rows, after using dropna to remove the rows that contain missing data, the final data was of 46,654 rows. Also, I decided to delete rather than filling because the number of missing data is small compared to the total size, and after dropping I would have around 46,000 rows, which is sufficient for analysis. Moreover, all the statistical tests work with real data. Moreover, if I replaced any missing value with average or something it might hide important variations and look different from the reality creating a false analysis. I couldn't know why people left certain questions unanswered. Maybe people who are very heavy or very light feel insecure to answer these questions.

In task 1, I performed calculations of the mean, median, standard deviation, Q1, and Q3 for the three variables: weight2, wtyrago, and htm3. The average age of the participants is around 54.8 years, and the median age is also 55 years. The data consists largely of the middle-aged and older population groups. This is important since the weights changes across various age groups. People tend to gain weight during their 30s and 50s and then start shedding it during old age. The average and median of weight were almost same for the current year and the past year (average and median of around 79 kg and 77 kg). The standard deviation is around 20 kg. Some people have got heavier, whereas others have lost weight and then some are the same. This is because people's lifestyles, level of stress, and eating habits are not the same. The average height of the population would be 169 cm, and the standard deviation would be around 10 cm. As the height of the adult population does not change so the standard deviation of height would be smaller than that of weight.

The bar chart of the means states that the average current weight and last year's weight are almost equal. This is a normal trend because adults normally do not experience many changes in weight. Overall, the data presented shows a realistic representation of the body characteristics of adults. So, the average weight remains steady while the variability is large, and the height that follows a predictable pattern.

For task 2, I made a new variable:weight_change = weight2 - wtyrago

Here a positive result represents a gain, and a negative result represents a loss. I performed the correlation calculations for the change in weight and the three variables, which are current weight, last year's weight, and age

My results were:

Correlation with Weight2 =+0.02

Correlation with wtyrago = –0.32

Correlation with age = -0.07

The strongest relationship is the relationship between weight_change and wtyrago. The negative sign (-0.32) indicates that individuals who were heavier last year lost weight, and individuals who were lighter got some weight. Maybe the heavier people were trying to lose their weight whereas the people who were lighter tried to gain some weight.The relationship between weight_change and current.weight (0.02) is nothing. As two people could have the same current.weight but very different weight_change experiences. The graph of weight_change vs current.weight showed random points with no specific pattern. The correlation is slightly negative (-0.07) indicating that people tend to lose slightly more weight with age, but the difference is very small. This could be because people tend to lose muscle mass with age.

3.1 Weight difference between the male and female groups

Using the 'ttest_ind' function, I tested whether the weight_change of males (sex = 1), and females (sex = 2) differ significantly.

t=-3.7986

p = .0001457

The p value here is very small which indicating the difference is real. As the t-value is negative, it indicates male data points are slightly lower. Men lost a bit more weight than females. But I refrain from large claims since I do not have information concerning effect size.

3.2 Random Group A vs Group B

After shuffling rows and splitting the dataset into two random halves, the t-test revealed that:

t=-0.59

p = 0.55

The large p-value indicates the absence of a significant difference between any two random splits of the same data. This verifies the functionality of the code, and a random split should not have any real effect.

3.3 Repeating random splits 1000 times

To see the changes of p value I split randomly and ran the t test for 1000 times. The values of p were big that's why most of the bars in the graph on the left side. Big p values mean that the difference is random, and the groups are same. Some of the bars were higher that didn't mean that they were different, it happened because of random chance. Although there's no huge difference but sometimes the system displays small number of p by accident, that's why it looks like that. This thing is called false positive.

3.4 Ratio difference between males and females

I did:

weight_height_ratio = weight2 / htm3

The t-test result for males and females is:

t = 57.73

p = 0.0

The large t-statistic is a result of two factors: the true differences in body composition and the large sample size. With so many rows of data, even a moderate difference is statistically huge. It helped me to understand that there are many differences between statistical significance and practical significance.

3.5 Repeating Weight Change Test

The repeated test produced the same result as section 3.1:

t=-3.7986

p = .0001457

This shows my consistency and reproducibility.

Data obtained from BRFSS needs ethical consideration, as works for real people, real bodies, and real experiences. Although this is anonymous data, certain variables might allow someone from a small community to identify an individual’s age, height, and weights. In my case as a student, I tried to make sure that this did not happen. The data set only consists of male and female as the only two genders, excluding transgender and non-binary genders. This directly affects the diversity of the data collection and analysis. Additionally, “dropna” is used to exclude those who didn’t provide answers to various questions, might be stigma, language issues, and other issues. The variables related to body are very sensitive, and various issues are likely to be raised, especially while comparing both genders. Various other factors, including income, culture, stress, and drugs, affect weight variation. Moreover, I included human dignity, truth, and transparency. Every process undertaken in this project was described to avoid any misunderstanding against the data obtained. This project taught me the full process of analysing data with Python, from preparation to summarization, visualizations of trends, computing correlations, t-testing, and handling ethics. Some of the important observations were the targeted group being seniors, steady mean weights over the course of a year, and the previous year’s weights on the variation. Overall, the health data was analysed in a clear, respectful, and ethical manner.